

Vector Calculus

Solutions to Problem Sheet 5

1. (a) $\nabla f = (2x - 3x^2y^2)\mathbf{i} + (4y^3 - 2x^3y)\mathbf{j}.$
 (b) $\nabla f = 4xye^{-2x^2y}\mathbf{i} + (1 + 2x^2e^{-2x^2y})\mathbf{j}.$
 (c) $\nabla f = \frac{y^2}{z^3}\mathbf{i} + \frac{2xy}{z^3}\mathbf{j} - \frac{3xy^2}{z^4}\mathbf{k}.$
 (d) $\nabla f = y \cos(yz)\mathbf{i} + [x \cos(yz) - xyz \sin(yz)]\mathbf{j} - xy^2 \sin(yz)\mathbf{k}.$

2. (a) $\phi = 2x^2y + 3xyz + yz^3$

$$\begin{aligned} \text{grad } \phi &= \nabla \phi = (4xy + 3yz)\mathbf{i} + (2x^2 + 3xz + z^3)\mathbf{j} + (3xy + 3yz^2)\mathbf{k} \\ &= 2\mathbf{i} - 2\mathbf{j} + 12\mathbf{k} \quad \text{at the point } (1, 2, -1). \end{aligned}$$

$$\begin{aligned} \text{div}(\text{grad } \phi) &= \nabla^2 \phi = \frac{\partial}{\partial x}(4xy + 3yz) + \frac{\partial}{\partial y}(2x^2 + 3xz + z^3) + \frac{\partial}{\partial z}(3xy + 3yz^2) \\ &= 4y + 0 + 6yz \\ &= -4 \quad \text{at } (1, 2, -1). \end{aligned}$$

- (b) $\phi = x^2z + \exp(y/x)$

$$\begin{aligned} \text{grad } \phi &= \left(2xz - \frac{y}{x^2} \exp(y/x)\right)\mathbf{i} + \frac{1}{x} \exp\left(\frac{y}{x}\right)\mathbf{j} + x^2\mathbf{k} \\ &= (-2 - 2e^2)\mathbf{i} + e^2\mathbf{j} + \mathbf{k} \quad \text{at the point } (1, 2, -1) \end{aligned}$$

$$\begin{aligned} \text{div}(\text{grad } \phi) &= 2z + \frac{2y}{x^3} \exp(y/x) + \frac{y^2}{x^4} \exp(y/x) + \frac{1}{x^2} \exp(y/x) \\ &= -2 + 9e^2 \quad \text{at } (1, 2, -1). \end{aligned}$$

3. $\mathbf{v} = 2xy\mathbf{i} + z^2\mathbf{j} + xyz\mathbf{k}$

$$\text{div } \mathbf{v} = \nabla \cdot \mathbf{v} = \frac{\partial}{\partial x}(2xy) + \frac{\partial}{\partial y}(z^2) + \frac{\partial}{\partial z}(xyz) = 2y + xy$$

$$\begin{aligned} \text{curl } \mathbf{v} &= \nabla \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy & z^2 & xyz \end{vmatrix} \\ &= \mathbf{i}(xz - 2z) - \mathbf{j}(yz - 0) + \mathbf{k}(0 - 2x) \\ &= (xz - 2z)\mathbf{i} - yz\mathbf{j} - 2x\mathbf{k} \end{aligned}$$

4. (a)

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(xz) + \frac{\partial}{\partial y}(yz) + \frac{\partial}{\partial z}(xy) = 2z.$$

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ xz & yz & xy \end{vmatrix} = (x-y)\mathbf{i} + (x-y)\mathbf{j} = (x-y)(\mathbf{i} + \mathbf{j}).$$

(b)

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(3x^2y) + \frac{\partial}{\partial y}(2xz^3) + \frac{\partial}{\partial z}(y^4) = 6xy.$$

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ 3x^2y & 2xz^3 & y^4 \end{vmatrix} = (4y^3 - 6xz^2)\mathbf{i} + (2z^3 - 3x^2)\mathbf{k}.$$

(c)

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(xe^{-z}) + \frac{\partial}{\partial y}(4yz^2) + \frac{\partial}{\partial z}(3ye^{-z}) = e^{-z} + 4z^2 - 3ye^{-z}.$$

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ xe^{-z} & 4yz^2 & 3ye^{-z} \end{vmatrix} = (3e^{-z} - 8yz)\mathbf{i} - xe^{-z}\mathbf{j}.$$

(d)

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^2 \sin(yz)) + \frac{\partial}{\partial y}(z \cos(xz^3)) + \frac{\partial}{\partial z}(ye^{5xy}) = 2x \sin(yz).$$

$$\begin{aligned} \nabla \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ x^2 \sin(yz) & z \cos(xz^3) & ye^{5xy} \end{vmatrix}, \\ &= [e^{5xy} + 5xye^{5xy} - \cos(xz^3) + 3xz^3 \sin(xz^3)]\mathbf{i} + [x^2y \cos(yz) - 5y^2e^{5xy}]\mathbf{j} \\ &\quad - [z^4 \sin(xz^3) + x^2z \cos(yz)]\mathbf{k}. \end{aligned}$$

5. (a)

$$\operatorname{div} \mathbf{r} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3.$$

(b)

$$\operatorname{curl} \mathbf{r} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ x & y & z \end{vmatrix} = 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} = \mathbf{0}.$$

(c)

$$\mathbf{a} \times \nabla = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ \partial_x & \partial_y & \partial_z \end{vmatrix} = (a_2\partial_z - a_3\partial_y)\mathbf{i} + (a_3\partial_x - a_1\partial_z)\mathbf{j} + (a_1\partial_y - a_2\partial_x)\mathbf{k}.$$

$$\begin{aligned} (\mathbf{a} \times \nabla) \times \mathbf{r} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_2\partial_z - a_3\partial_y & a_3\partial_x - a_1\partial_z & a_1\partial_y - a_2\partial_x \\ x & y & z \end{vmatrix}, \\ &= -2a_1\mathbf{i} - 2a_2\mathbf{j} - 2a_3\mathbf{k} = -2\mathbf{a}. \end{aligned}$$

(d)

$$\begin{aligned} \mathbf{a} \times \mathbf{r} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ x & y & z \end{vmatrix} = (a_2z - a_3y)\mathbf{i} + (a_3x - a_1z)\mathbf{j} + (a_1y - a_2x)\mathbf{k}. \\ \nabla \cdot (\mathbf{a} \times \mathbf{r}) &= \frac{\partial}{\partial x}(a_2z - a_3y) + \frac{\partial}{\partial y}(a_3x - a_1z) + \frac{\partial}{\partial z}(a_1y - a_2x) = 0. \end{aligned}$$

(e)

$$(\mathbf{r} \cdot \mathbf{r})\mathbf{a} = (x^2 + y^2 + z^2)\mathbf{a}.$$

$$\begin{aligned} \nabla \times [(\mathbf{r} \cdot \mathbf{r})\mathbf{a}] &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ (x^2 + y^2 + z^2)a_1 & (x^2 + y^2 + z^2)a_2 & (x^2 + y^2 + z^2)a_3 \end{vmatrix}, \\ &= (2ya_3 - 2za_2)\mathbf{i} + (2za_1 - 2xa_3)\mathbf{j} + (2xa_2 - 2ya_1)\mathbf{k}. \\ \mathbf{r} \times \mathbf{a} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & z \\ a_1 & a_2 & a_3 \end{vmatrix} = (ya_3 - za_2)\mathbf{i} + (za_1 - xa_3)\mathbf{j} + (xa_2 - ya_1)\mathbf{k}. \end{aligned}$$

Hence, LHS = RHS and $\nabla \times [(\mathbf{r} \cdot \mathbf{r})\mathbf{a}] = 2(\mathbf{r} \times \mathbf{a})$.

(f)

$$\begin{aligned} \nabla \cdot [(\mathbf{r} \cdot \mathbf{r})\mathbf{a}] &= \frac{\partial}{\partial x}((x^2 + y^2 + z^2)a_1) + \frac{\partial}{\partial y}((x^2 + y^2 + z^2)a_2) + \frac{\partial}{\partial z}((x^2 + y^2 + z^2)a_3), \\ &= 2xa_1 + 2ya_2 + 2za_3 = 2(\mathbf{r} \cdot \mathbf{a}). \end{aligned}$$

$$6. \mathbf{A} = (2x^2, -3yz, xz^2) \quad \phi = 2z - x^3y$$

$$\nabla\phi = (-3x^2y, -x^3, 2)$$

$$\begin{aligned} \mathbf{A} \cdot \nabla\phi &= -6x^4y + 3x^3yz + 2xz^2 \\ &= 5 \quad \text{at} \quad (1, -1, 1) \end{aligned}$$

$$\begin{aligned} \mathbf{A} \times \nabla\phi &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2x^2 & -3yz & xz^2 \\ -3x^2y & -x^3 & 2 \end{vmatrix}, \\ &= (-6yz + x^4z^2, -3x^3yz^2 - 4x^2, -2x^5 - 9x^2y^2z) \\ &= (7, -1, -11) \quad \text{at} \quad (1, -1, 1) \end{aligned}$$

7. $\phi = \tan^{-1} \left(\frac{y}{x} \right)$. Now, let $u = \frac{y}{x}$. Then

$$\begin{aligned} \frac{\partial}{\partial x} \tan^{-1} \left(\frac{y}{x} \right) &= \frac{\partial}{\partial x} \tan^{-1} u, \\ &= \frac{1}{1+u^2} \frac{\partial u}{\partial x}, \\ &= \frac{x^2}{x^2+y^2} \left(-\frac{y}{x^2} \right) \\ &= -\frac{y}{x^2+y^2}. \end{aligned}$$

Similarly,

$$\frac{\partial}{\partial y} \tan^{-1} \left(\frac{y}{x} \right) = \frac{x}{x^2+y^2}.$$

Hence, at $(2, -2)$, $\nabla\phi = \left(\frac{1}{4}, \frac{1}{4} \right)$. The directional derivative in the direction of $\mathbf{u} = (1, -3)$ is

$$\frac{\mathbf{u}}{|\mathbf{u}|} \cdot \nabla\phi = \frac{\frac{1}{4}(1, 1) \cdot (1, -3)}{\sqrt{10}} = -\frac{1}{2\sqrt{10}}.$$

8. $\phi = xy^2z + 4yz^2$

$$\begin{aligned} \nabla\phi &= (y^2z, 2xyz + 4z^2, xy^2 + 8yz) \\ &= (-1, 8, -10) \quad \text{at} \quad (-2, 1, -1). \end{aligned}$$

Directional derivative in direction of $\mathbf{u} = (-1, 2, -2)$ is

$$\frac{\mathbf{u}}{|\mathbf{u}|} \cdot \nabla\phi = \frac{(-1, 2, -2) \cdot (-1, 8, -10)}{3} = \frac{37}{3}.$$

9. (a)

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x + y^2 + az & bxy - 4y - z & 3x + cy + 2z \end{vmatrix} = (c+1, a-3, by-2y).$$

Hence $\nabla \times \mathbf{A} = (c+1, a-3, by-2y) = \mathbf{0}$

when $c = -1$, $a = 3$ and $b = 2$.

(b)

$$\begin{aligned} \nabla \times \mathbf{A} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix}, \\ &= \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \mathbf{i} - \left(\frac{\partial A_3}{\partial x} - \frac{\partial A_1}{\partial z} \right) \mathbf{j} + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \mathbf{k} \end{aligned}$$

$$\begin{aligned}
\nabla \cdot (\nabla \times \mathbf{A}) &= \frac{\partial}{\partial x} \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) - \frac{\partial}{\partial y} \left(\frac{\partial A_3}{\partial x} - \frac{\partial A_1}{\partial z} \right) + \frac{\partial}{\partial z} \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right), \\
&= \frac{\partial^2 A_3}{\partial x \partial y} - \frac{\partial^2 A_2}{\partial x \partial z} - \frac{\partial^2 A_3}{\partial x \partial y} + \frac{\partial^2 A_1}{\partial y \partial z} + \frac{\partial^2 A_2}{\partial x \partial z} - \frac{\partial^2 A_1}{\partial y \partial z}, \\
&= 0.
\end{aligned}$$

$$\begin{aligned}
10. \quad \nabla \cdot \mathbf{v} &= (2xyz) + \left(x \log y + xy \frac{1}{y} \right) + (e^x \cos z) \\
&= 2xyz + x \log y + x + e^x \cos z.
\end{aligned}$$